

Computer Aided Engineering Graphics (2025-26)

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Chapter 1

Engineering_Graphics

Chapter 2

Projection of Lines

2.1 Introduction

Projection of lines is a fundamental topic in engineering graphics. It involves representing a line in different planes (such as the horizontal plane and vertical plane) to understand its true length and inclination.

2.2 Objectives

- Understand the concept of projection of lines.
- Learn to draw projections of lines inclined to one or both reference planes.
- Solve typical problems related to projection of lines.

2.3 Basic Concepts

- Projection: The representation of an object on a plane.
- Reference Planes: Usually the Horizontal Plane (HP) and Vertical Plane (VP).
- True Length: The actual length of the line in 3D space.
- Apparent Length: The length of the line as seen in the projection.

2.4 Types of Line Positions

2.4.1 Line parallel to both HP and VP.

- In this scenario, the projections on both the V.P. (front view) and H.P. (top view) will display the true length of the line.

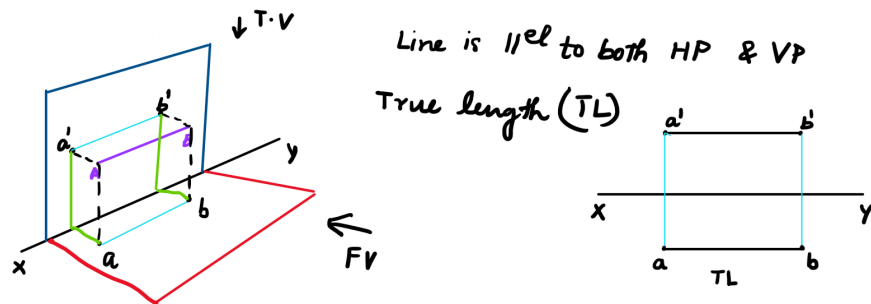


Figure 2.1: Projection of a line parallel to both HP and VP

2.4.2 Line parallel to one plane and perpendicular to the other.

- When the line is parallel to the V.P., its projection on the V.P. (front view) shows the true length, while its projection on the H.P. (top view) appears as a point.
- When the line is parallel to the H.P., its projection on the H.P. (top view) shows the true length, while its projection on the V.P. (front view) appears as a point.

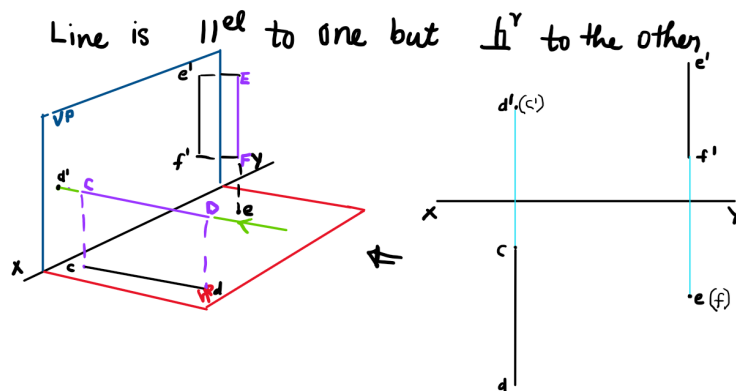
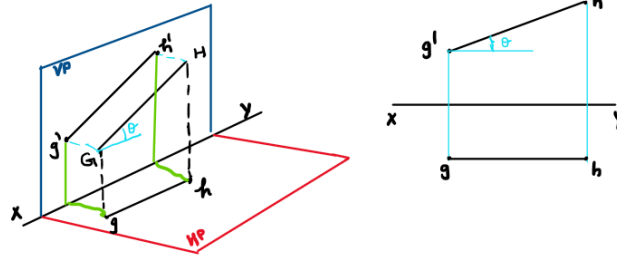


Figure 2.2: Projection of a line parallel to one plane and perpendicular to the other

2.4.3 Line parallel to one plane and inclined to the other.

- If the line is parallel to the V.P. and inclined to the H.P., the front view (on V.P.) will show the true length (TL), while the top view (on H.P.) will display an apparent length (shorter than the TL).
- If the line is parallel to the H.P. and inclined to the V.P., the top view (on H.P.) will show the true length (TL), while the front view (on V.P.) will display an apparent length (shorter than the TL).

Line is $\parallel$ to VP and inclined to HP



Line $\parallel$ to HP and Inclined to VP

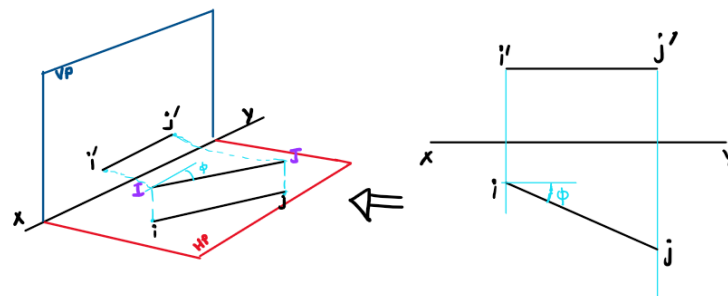


Figure 2.3: Projection of a line parallel to one plane and inclined to the other

2.4.4 Line inclined to both HP and VP.

Line inclined to both HP and VP

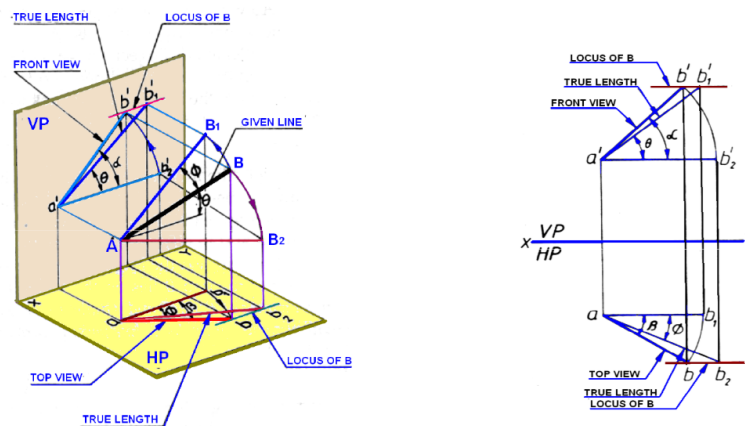


Figure 2.4: Projection of a line inclined to both HP and VP

- When a line is inclined to both the HP and VP, neither projection (front view or top view) shows the true length. Both projections appear shorter than the actual length of the line.
- The true length and true inclinations can be determined using auxiliary views or by rotating the projections.

2.4.4.1 Rotating Line AB to AB (Making it Parallel to VP)

To determine the true length and inclination of a line inclined to both HP and VP, we often use the rotation method. In this method, the line AB is rotated about one of its ends (say, point A) until it becomes parallel to the VP, resulting in a new position AB .

- Front View (Projection on VP): When AB is rotated to AB (parallel to VP), the front view shows the true length of the line.
- Top View (Projection on HP): The top view during this rotation will show the apparent length, which is shorter than the true length.

2.4.4.2 Rotating Line AB to AB (Making it Parallel to HP)

Similarly, to find the true length and inclination of a line, the line AB can be rotated about one of its ends (say, point A) until it becomes parallel to the HP, resulting in a new position AB .

- Top View (Projection on HP): When AB is rotated to AB (parallel to HP), the top view shows the true length of the line.
- Front View (Projection on VP): The front view during this rotation will show the apparent length, which is shorter than the true length.

[Image reference: IIT Guwahati Engineering Graphics](#)

Chapter 3

Projection of Planes

3.1 Introduction

- A plane is a two-dimensional surface that has only length and breadth; its thickness is considered negligible.
- In engineering graphics, planes are depicted using various geometric shapes, including:
 - Squares
 - Rectangles
 - Circles
 - Pentagons
 - Hexagons
 - Other polygons
- These plane figures form the foundation for understanding and visualizing the projection of objects onto different reference planes.

3.2 Typical Problem Statements

- What is usually asked in problems?
 - To draw the projections:
 - * Front View (F.V.)
 - * Top View (T.V.)
 - * Side View (S.V.)
- What information will be given in the problem?
 - The type/description of the plane figure (triangle, square, hexagon, etc.).
 - Its position with respect to Horizontal Plane (H.P.) and Vertical Plane (V.P.).
- How will the position with H.P. & V.P. be described?
 - The inclination of the surface/ lamina of the plane with one of the reference planes is given.

- The position/inclination of one of its corners/edges with respect to the other reference plane.
- This means the plane, we consider, is usually inclined to both reference planes.

3.3 Stages for Drawing Projections When a Plane is Inclined

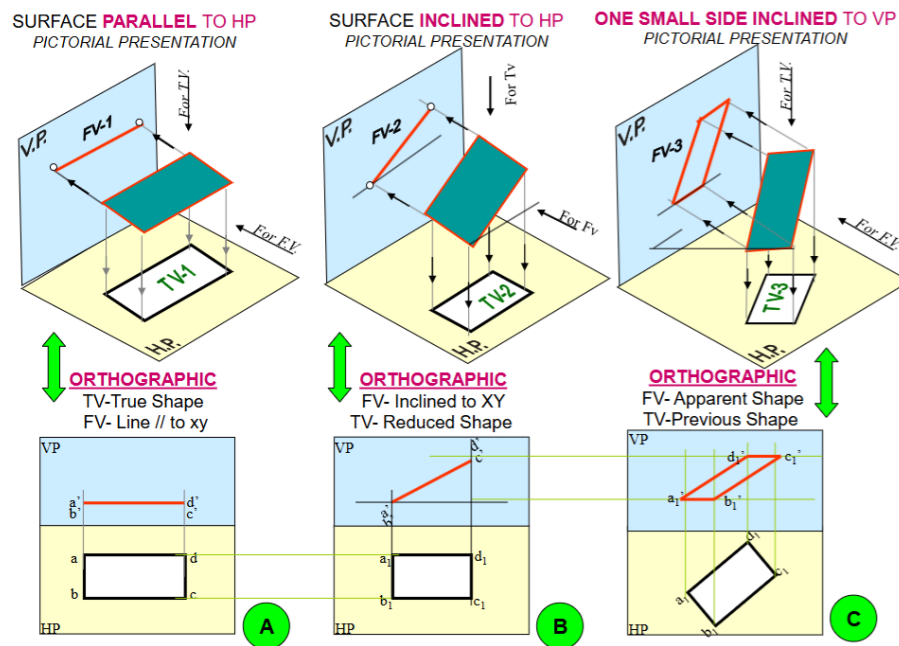


Figure 3.1: Stages of Drawing Projections for an Inclined Plane.

[image reference](#)

The process starts by assuming an initial simple position, then introducing the surface inclination, and finally introducing the side/edge inclination.

- Initial Position: Surface Parallel to one Reference Plane
 - Assumption: The plane figure is initially assumed to be parallel to one of the reference planes (either HP or VP). This allows one of its views to show its true shape.
 - Views:
 - * If the surface is assumed parallel to HP, the Top View (TV) will show the true shape of the plane figure. The Front View (FV) will appear as a line parallel to the XY line.
 - * If the surface is assumed parallel to VP, the Front View (FV) will show the true shape of the plane figure. The Top View (TV) will appear as a line parallel to the XY line.

- Surface Inclination
 - Description: The inclination of the plane's surface (lamina) with one of the reference planes is introduced in this step.
 - Transformation: The view that was a line in the previous step (e.g., FV if the surface was parallel to HP) will now be drawn inclined to the XY line at the given angle. The other view (e.g., TV) will show a reduced shape.
- Edge/Side Inclination with the Other Reference Plane
 - Description: The inclination of one of the plane's edges (or a specific feature like a diameter or diagonal) with the other reference plane is introduced. This leads to a case where the object is inclined to both reference planes.
 - Transformation: The view from the previous step that showed the "reduced shape" (e.g., TV in Problem 1) is now redrawn such that the specified edge or feature makes the given angle with the XY line (representing the other reference plane). The final views will show apparent shapes.

i How to Approach a Projection Problem

Read the problem and answer the following questions:

1. Surface inclined to which plane? "H.P. or V.P."
2. Assumption for initial position? "Parallel to H.P or V.P."
3. So which view will show the true shape? ; "Top view or Front View"
4. How to draw the edge/ corner of the plane which rests on one of the planes?
 "If an edge of a plane rests on plane draw the view in step - 1, with the edge perpendicular to the XY line. In